1 Solve the simultaneous equations

$$x^2 + y^2 = 13$$
$$x = y - 5$$

$(y-5)^2 + y^2 = 13$

$(y-5)(y-5) + y^2 = 13$

$y^2 - 5y - 5y + 25 + y^2 = 13$

$2y^2 - 10y + 25 = 13$

$2y^2 - 10y + 12 = 0$

$y^2 - 5y + 6 = 0$

$(y-3)(y-2) = 0$

$y = 3$ $\quad$ $y = 2$

$x = (3) - 5$ $\quad$ $x = (2) - 5$

$= -2$ $\quad$ $= -3$

$x =$ -2 or -3

$y =$ 3 or 2

(Total for question 1 is 5 marks)

2 Solve the simultaneous equations

$$x^2 + y^2 = 17$$
$$y = x - 3$$

$x^2 + (x-3)^2 = 17$

$x^2 + (x-3)(x-3) = 17$

$x^2 + x^2 - 3x - 3x + 9 = 17$

$2x^2 - 6x + 9 = 17$

$2x^2 - 6x - 8 = 0$

$x^2 - 3x - 4 = 0$

$(x-4)(x+1) = 0$

$x = 4 \quad x = -1$

$y = (4) - 3 \qquad y = (-1) - 3$

$= 1 \qquad = -4$

$x =$ 4 or −1

$y =$ 1 or −4

(Total for question 2 is 5 marks)

3 Solve the simultaneous equations

$$x^2 + y^2 = 34$$
$$x - y = 2$$

$x = 2 + y$

$(2+y)^2 + y^2 = 34$

$(2+y)(2+y) + y^2 = 34$

$4 + 2y + 2y + y^2 + y^2 = 34$

$2y^2 + 4y + 4 = 34$

$2y^2 + 4y - 30 = 0$

$y^2 + 2y - 15 = 0$

$(y+5)(y-3) = 0$

$y = -5 \quad y = 3$

$x = 2 + (-5) \quad x = 2 + (3)$

$= -3 \quad = 5$

$x =$ -3 or 5

$y =$ -5 or 3

(Total for question 3 is 5 marks)

4 Solve the simultaneous equations

$$x^2 + y^2 = 20$$
$$3x = 2 - y$$

$3x + y = 2$

$y = 2 - 3x$

$x^2 + (2-3x)^2 = 20$

$x^2 + (2-3x)(2-3x) = 20$

$x^2 + 4 - 6x - 6x + 9x^2 = 20$

$10x^2 - 12x + 4 = 20$

$10x^2 - 12x - 16 = 0$

$5x^2 - 6x - 8 = 0$

$(5x + 4)(x - 2) = 0$

$x = \frac{-4}{5}$ $\quad$ $x = 2$

$y = 2 - 3\left(\frac{-4}{5}\right)$ $\quad$ $y = 2 - 3(2)$

$= 2 + \frac{12}{5}$ $\quad$ $= -4$

$= \frac{22}{5}$

$x =$ $-4/5$ or 2

$y =$ $22/5$ or -4

(Total for question 4 is 5 marks)

5 Solve the simultaneous equations

$$x^2 + y^2 = 41$$
$$y = 2x - 3$$

$x^2 + (2x-3)^2 = 41$

$x^2 + (2x-3)(2x-3) = 41$

$x^2 + 4x^2 - 6x - 6x + 9 = 41$

$5x^2 - 12x + 9 = 41$

$5x^2 - 12x - 32 = 0$

$(5x+8)(x-4) = 0$

$x = -\frac{8}{5}$ $\quad$ $x = 4$

$y = 2\left(-\frac{8}{5}\right) - 3$ $\quad$ $y = 2(4) - 3$

$= -\frac{16}{5} - 3$ $\quad$ $= 5$

$= -\frac{31}{5}$

$x =$ $-8/5$ or 4

$y =$ $-31/5$ or 5

(Total for question 5 is 5 marks)

6 Solve the simultaneous equations

$$x^2 + y^2 = 20$$
$$2x + y = 3$$
$$y = 3 - 2x$$

$$x^2 + (3 - 2x)^2 = 20$$

$$x^2 + (3-2x)(3-2x) = 20$$

$$x^2 + 9 - 6x - 6x + 4x^2 = 20$$

$$5x^2 - 12x + 9 = 20$$

$$5x^2 - 12x - 11 = 0$$

This will not factorise.

$a = 5 \quad b = -12 \quad c = -11$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-12) \pm \sqrt{(-12)^2 - 4(5)(-11)}}{2(5)}$$

$x = 3.11$ (3sf) or -0.708 (3sf)

$y = 3 - 2(\text{"}3.11\text{"})$ $\quad$ $y = 3 - 2(\text{"}-0.708\text{"})$

$= -3.22$ (3sf) $\quad$ $= 4.42$ 3sf

$x =$ 3.11 or −0.708

$y =$ −3.22 or 4.42

(Total for question 6 is 5 marks)

7 Solve the simultaneous equations

$$x^2 + y^2 = 27$$
$$2x - y = 3$$

$2x = 3 + y$

$2x - 3 = y$

$x^2 + (2x-3)^2 = 27$

$x^2 + (2x-3)(2x-3) = 27$

$x^2 + 4x^2 - 6x - 6x + 9 = 27$

$5x^2 - 12x + 9 = 27$

$5x^2 - 12x - 18 = 0$

$a = 5 \quad b = -12 \quad c = -18$

$$x = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(5)(-18)}}{2(5)}$$

$= \underline{\underline{3.44}}$ (3sf) or $\underline{\underline{-1.04}}$ (3sf)

$y = 2(\text{Ans}) - 3$ $\qquad$ $y = 2(\text{Ans}) - 3$

$= \underline{\underline{3.89}}$ (3sf) $\qquad$ $= -5.09$ (3sf)

$x =$ 3.44 or −1.04

$y =$ 3.89 or −5.09

(Total for question 7 is 5 marks)

8 Solve algebraically the simultaneous equations

$$x^2 - 3y^2 = 13$$
$$2x + 3y = 4$$

$2x + 3y = 4$

$2x = 4 - 3y$

$x = \frac{4 - 3y}{2}$

$\left(\frac{4 - 3y}{2}\right)^2 - 3y^2 = 13$

$\left(\frac{4 - 3y}{2}\right)\left(\frac{4 - 3y}{2}\right) - 3y^2 = 13$

$\frac{16 - 12y - 12y + 9y^2}{4} - 3y^2 = 13$

$\frac{16 - 24y + 9y^2}{4} - 3y^2 = 13$

$16 - 24y + 9y^2 - 12y^2 = 52$

$-3y^2 - 24y + 16 = 52$

$-3y^2 - 24y - 36 = 0$

$3y^2 + 24y + 36 = 0$

$y^2 + 8y + 12 = 0$

$(y + 2)(y + 6) = 0$

$y = -2 \quad y = -6$

$x = \frac{4 - 3(-2)}{2}$

$= 5$

$x = \frac{4 - 3(-6)}{2}$

$= 11$

$x = 5, \ y = -2$ or $x = 11, \ y = -6$

(Total for question is 5 marks)

9 Solve algebraically the simultaneous equations

$$2x^2 - y^2 = 14$$
$$3x + 2y = 3$$

$3x + 2y = 3$

$3x = 3 - 2y$

$x = \frac{3 - 2y}{3}$

$2\left(\frac{3-2y}{3}\right)^2 - y^2 = 14$

$2\left(\frac{3-2y}{3}\right)\left(\frac{3-2y}{3}\right) - y^2 = 14$

$2\left(\frac{9 - 6y - 6y + 4y^2}{9}\right) - y^2 = 14$

$2\left(\frac{9 - 12y + 4y^2}{9}\right) - y^2 = 14$

$\frac{18 - 24y + 8y^2}{9} - y^2 = 14$

$18 - 24y + 8y^2 - 9y^2 = 126$

$-y^2 - 24y + 18 = 126$

$-y^2 - 24y - 108 = 0$

$y^2 + 24y + 108 = 0$

$(y + 6)(y + 18) = 0$

$y = -6 \qquad y = -18$

$x = \frac{3 - 2(-6)}{3}$

$= 5$

$x = \frac{3 - 2(-18)}{3}$

$= 13$

$x = 5$ and $y = -6$ or $x = 13$ and $y = -18$

(Total for question is 5 marks)